Introduction to AI.

→ Artifical Intelligence:

- Definition: It is the branch of Computer Science that develops machines or computer programs that can perform tasks that normally require human intelligence.

*Human Intelligence include:

- Learning
- Reasoning
- Problem Solving
- Decision making
- understanding language
- Recognizing objects.
- Planning.

* Goals of AI:

The main goal of AI is to develop systems that can perform intelligent tasks effectively.

1. Problem Solving: AI should be able to find solutions to problems.
2. Learning: AI should be able to improve its performance using experience or data.
3. Reasoning: AI should be able to use available information to reach conclusions.

4. Decision making : AI should select an appropriate action from different possibilities.

5. Perception: AI should be able to understand information obtained from the environment

6. Natural language understanding: AI should be able to understand and communicate using human language.

7. Planning : AI should be able to determine a sequence of actions needed to achieve a goal

* Applications:
- Healthcare
- Education
- E-Commerce
- Games
- Language

AI = Sense → Think → Decide → Act

* characteristics.
- Learning
- Problem Solving
- Reasoning
- Decision making
- Knowledge Representation
- Perception
- Goal oriented Behaviour

→ AI Problems :

- An AI problem is a problem that requires intelligent decision making, reasoning, searching, planning, learning or understanding to find a solution.

*Types of AI Problems

1. Search Problems
2. Planning Problems.
3. Knowledge and reasoning problems.
4. classification problem.
5. Prediction problems.

1. Search problems : The system needs to search through possible solutions.

Example : BFS
DFS
Best First Search
A*

2. Planning problem : The system must determine a sequence of actions to achieve a goal.

Example : Robotics.

- If we assign a work it plans and does the work

3. Knowledge and Reasoning problem :
The system must use available knowledge to reach conclusions.

4. Classification Problem.

- The System need to place something into a category.

5. Prediction Problem.

- The System predicts an outcome based on available information.

* Problem Solving Approaches in AI:

process of AI problem solving.

Problem
↓
initial state
↓
Goal state
↓
possible Actions.
↓
Search Solutions
↓
Select Suitable Solutions (Execute Solution)

Step 1: Define the Problem

clearly identify what needs to be solved

Step 2: Initial state

where are we starting.

Step 3: Goal state

what do we want to achieve.

Step 4: Actions

what possible actions we take.

Step 5: Search

The AI explores possible route.

Step 6: Select Solution

AI chooses an appropriate route based on problem requirement.

Step 6: Execute

The selected solution is carried out

Example :- Find a route between two locations.

A → B.

AI searches possible route.

⇒ AI Techniques :

- These are methods used to represent Knowledge, Solve problems, reason, learn, plan and make decisions.

1. Search Techniques.
2. Knowledge Representation.
3. Reasoning
4. Learning
5. Planning.

1. Search Techniques:

- Search means exploring possible solutions to find desired solutions.
- Uniformed search, don't use additional knowledge

Ex: BFS, DFS.

- Informed | Heuristic :

Uses additional information to guide it.

Ex: Best first search, A*

2. Knowledge Representation :: It means representing information in a form that a computer can understand and use for reasoning.

• AI systems needs knowledge to make intelligent decisions.

3. Reasoning:

• It is the process of using known information to reach a conclusion.

Types:- • Forward
• Backward
• Resolution based.

4. Learning:

• It is the ability of a system to improve its performance using experience or data

• It learn patterns.

5. Planning:

• It is determining a sequence of actions that will achieve a goal.

⇒ State Space Search.

• It is used to solve problems by exploring possible states.

• A state space is the set of all possible states that a system can be in problem solving.

• Before Solving
→ Define the problem
→ Analyze the problem
→ Identify one possible solution
→ choose the best solution.

• state space search → 4 Tuple (S, I, O, G)

S → set of states, I → Initial state

O → operations / Actions, process of moving from one state to another.

G → Goal state.

*8 puzzle problem:

• possible Actions : up, down, left, right

Initial state:

1	2	3
7		4
6	8	5

3x3 board
8 numbered tiles

Goal state

1	2	3
8		4
7	6	5

↓

move up

1	2	3
7	8	4
6	↑	5

→ move down

1	2	3
7	8	4
→	6	5

→ move left

1	2	3
	8	4
7	6	5

→ Goal st

1	2	3
8		4
7	6	5

* Problem as state space search:

• In AI problem can be represented as a state space the AI starts from initial state applies

the operation/actions and searches through different states until it reaches the goal state.

1. State: A state represent a current situation or condition of a problem.

2. Initial State: the initial state is the state where the problem starts.

   Initial state = starting point

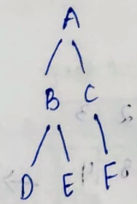
3. Goal State: the goal state is desired from final state that we want to achieve

- The AI searches for sequence of actions that changes the initial state into goal state

4. Operations / Actions:

   Operators are the actions that can be performed to change one state into another.

5. State space: collection of all possible states that can be reached by applying valid operations.

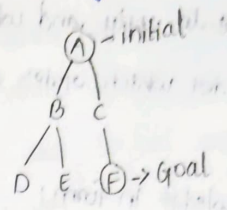


Here, A, B, C, D, E, F represent different possible states

6. State space representation:

- Initial space
- operations / Actions
- possible states
- Goal State
- path b/w states

Ex: Represent using Graph or Tree



A -> initial state
B, C, D, E -> possible states
F -> Goal
Branches -> operations / Actions
path :- A -> C -> F -> solution

⇒ Production Systems :

Definition : A production system is a AI technique used to represent knowledge and solve problems using rules.

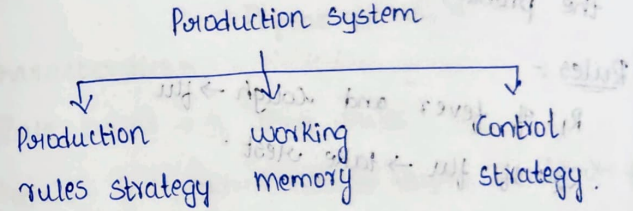
if Condition → Then Action / conclusion

Example :-

if it is raining

Then take an umbrella

* Components of Production Systems.



1. production rules

- Represent knowledge using if-then rules.
- Tell the systems what actions to take when a condition is satisfied.

2. working memory

- Stores the current facts / information about problem
- It is like temporary memory.

3. Control strategy.

- Decides which rule to apply and when
- It controls the order which rules are executed.
- Production rules:

It represents knowledge in form:

General form:

if <condition>

Then <action>

- Production rules tell the AI what to do when a condition is true.

Example:- if temp is high
then turn on fan.

4. working memory:

- It stores the current facts and information of the problem.

Rules:-

R_1: If fever and cough → flu

R_2: If flu → take rest.

Initial working memory

Fever = Yes } Since R_1 matches
Cough = Yes } fever + cough
↓
flu

working memory is updated.

Fever - Yes

Cough - Yes

Flu - Yes.

Now R_2 can be Applied

Flu
↓
Take rest

5. Control strategy:

It decide which rule should be selected and update
Sometimes several rules can be applied at the same time the control strategy chooses the appropriate rules.

Main steps

working memory
↓
Find application rules
↓
Select a rule
↓
Apply a rule
↓
update working memory.
↓
Repeat

* characterstics:-

- Rule based → If then rules
- Module - Independent these rules can be added remains modified easily
- Separation of Knowledge and Control
- Problem solving
- Handles changing information
- Supports inference - New facts from existing facts.

Ex:- R_1: If sky is cloudy
Then it will rain

R2 : If it may rain
Then carry umbrella

Initial, working memory
Sky is cloudy

Step 1 :
Sky is cloudy
↓
Apply R1
↓
It may rain

working memory becomes
Sky is cloudy
It may rain

Step 2 :- It may rain
↓
Apply R2
Carry umbrella

Final result
Sky is cloudy
↓
It may rain
↓
carry umbrella

⇒ Control Strategies of AI

Control strategy : It determines which production rule should be applied when it should be applied and in what order to solve problems.

- It decides how rules are applied to solve problem.

* Types of control strategy.

1. Forward Reasoning
2. Backward Reasoning.
3. Selection of Rules
4. conflict Resolution

1. Forward Reasoning :- It is a data driven reasoning method. It starts with known facts and applies rules to drive new facts until the goal is reached.

Facts → Rules → New facts → Goal

2. Backward Reasoning :- It is a goal driven reasoning method. It start with goal and works backwa to find the facts required to prove goal.

Goal → Rules → Required facts.

Ex :- Goal :
carry umbrella

Rules :
R1 : If cloudy → may rain
R2 : If may rain - Carry umbrella.

working :-
Carry umbrella
↑ R2
may rain
↑ R1
cloudy

Now, system checks whether clou
is a known fact
If cloudy is known goal is appro
Goal → fact

3. Selection of Rules

- Rule selection is the process of choosing an appropriate rule from the rules that can currently be applied.
- More than one rule is applicable. The system must decide which rule to apply.

Ex:- working memory
- fever
- cough

Rules:-

R_1: If fever → Take medicine

R_2: If fever and cough → consult doctor.

Both R_1 and R_2 can be applied.

So the system asks.

which rule should be apply first:

*selection factors:-

1. Priority → choose with high priority.
2. Specificity → choose the rule with more specific conditions.
3. Recency → Prefer rules using recently added information.
4. Rule order → choose accordingly to the order in which rules are listed.

5 Problem requirements: Select the rule most suitable for the current problem.

6. Conflict resolution:-

Definition: when 2 or more rules are applicable of the same time a conflict occurs.

- Conflict resolution is the process of deciding which rule should be applied.

Example:-

facts:
- fever
- cough

Rules:-

R_1: If fever: Take medicine.

R_2: If fever and cough → Consult doctor both

Both rules are applicable.

Therefore,

fever + cough
↓
$R_1 + R_2$
↓
Conflict resolution
↓
Select one rule.

Methods of conflict resolution

- Priority → higher priority
- Specificity → more specific conditions.
- Recency → must recently added or updated information

- Rule order:

If rules have the same priority, the system select the rule that appears first in the rule list.

⇒ uninformed search using BFS and DFS.

* Definition: uninformed search in an AI search technique that searches for a solution without using heuristic or additional knowledge about the problem. It uses only the information given in the problem.

- The two important uninformed search methods are:

1. Breadth First Search (BFS)
2. Depth First Search (DFS)

1. Breadth First Search (BFS).

- BFS explores the search tree level by level. It visits all nodes at one level before moving to the next level.

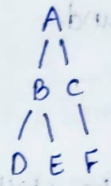
working:

- Start from the initial node.
- Visit all its immediate children.
- Then visit the children of those nodes.
- Continue until the goal is reached
- BFS uses a queue (FIFO)

Algorithm:

1. Insert the initial node into the queue.
2. Remove the first node from the queue.
3. check whether it is the goal.
4. If not, add its unvisited children to the queue.
5. Repeat until the goal is found or the queue is empty.

Example:



BFS order:

A → B → C → D → E → F

If E is the goal, BFS finds E after searching level by level

Advantages:

- Finds the shortest path when all step costs are equal
- complete if a solution exists at a finite depth.
- Easy to understand and implement

Disadvantages:

- Requires more memory
- Can be slow for large search spaces.
- stores many nodes in the queue.

2. Depth First Search (DFS).

- DFS explores a path as deeply as possible, before backtracking and trying another path.

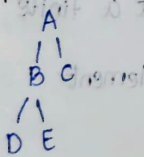
Working:

- Start from the initial node.
- Select one child and continue deeper.
- When no further child is available, backtrack.
- Continue until the goal is found.
- DFS uses a stack (LIFO).

Algorithm:

1. Insert the initial node into the stack.
2. Remove the top node.
3. Check whether it is the goal.
4. If not, add its unvisited children to the stack.
5. Continue until the goal is found or the stack becomes empty.

Example:



One possible DFS order:

A → B → D → E → C

If D is the goal, DFS finds it quickly by going deep through B.

Advantages:

- Requires less memory than BFS.
- Simple to implement.
- useful when the solution is deep in the search tree.

Disadvantages:

- Does not guarantee the shortest path.
- Can go very deep in an unproductive path.
- May not find a solution in an infinite search space.

BFS vs DFS

BFS	DFS
• Searches level by level	• Searches depth-wise
• Uses Queue	• Uses stack
• FIFO	• LIFO
• Shortest path for equal costs	• Shortest path not guaranteed
• Requires more memory	• Requires less memory.

⇒ Heuristic Search :-

- Heuristic search is an informed search technique used in AI to find a solution efficiently by using additional knowledge about the problem.
- It uses an estimate to decide which path is more promising.

Heuristic: A heuristic is a rule or method that helps us choose the best or most promising path toward the goal

Example:

In finding a route to a city, the straight-line distance to the destination can be used as a heuristic

*heuristic function:

- A heuristic function h(n) estimates the cost of distance from the current node n to the goal.
- h(n) = estimated cost from node n to the goal.
- Smaller h(n) generally means the node is closer to the goal

Example:

If the estimated distance from A to the goal is 10km:

$$h(A) = 10$$

* Evaluation function:

- An evaluation function is used to decide which node should be explored next.

In A* search:

$$f(n) = g(n) + h(n)$$

where:

- g(n) = actual cost from the start node to n
- h(n) = estimated cost from n to the goal.
- f(n) = total estimated cost of the solution

The node with the lowest f(n) is selected first.

* Informed search

Informed search uses heuristic knowledge to guide the search toward the goal.

Examples:

- Greedy Best-first search
- A* search

Advantages:

- usually faster than uninformed search.
- Explores fewer unnecessary nodes.
- uses problem-specific knowledge

* Uninformed search:

- uninformed search does not use heuristic knowledge. It searches using only the information available from the problem.

Examples:

- Breadth First Search (BFS)
- Depth First Search (DFS)
- uniform cost search

Advantages:

- simple to implement
- Does not require heuristic information

Disadvantages:

- May explore many unnecessary nodes.
- Can require more time and memory.

Informed Search	Uninformed Search
• uses heuristic knowledge	• Does not use heuristic knowledge
• more goal-directed	• Less goal-directed
• usually more efficient	• may explore many nodes
• Examples: A*, Greedy	• Examples: BFS, DFS

⇒ Problem Characteristics in AI:

Problem Characteristics describe the nature & difficulty of a problem. They help us choose the right AI technique to solve it.

1. Decomposability:

A complex problem can be devided into smaller, simpler problems.

Example: planning a trip can be divided into transport, hotel, food, etc.

2. Nature of Solution Steps:

It tells whether the solution requires a ~~(of an action can be predicted accurately)~~ fixed sequence of steps or different possible steps.

3. Predictability:

It shows ~~whether~~ whether the result of an action can be predicted accurately.

4. Importance of Knowledge:

Some problems need a large amount of domain knowledge to find a solution.

Example: "Medical diagnosis requires medical knowledge."

5. Role of Interaction:

It describes whether interaction with the user, environment, or other agents is needed while solving the problem.

Example: A chatbot interacts with user to understand and answer questions.

In short: Problem characteristics help an AI system determine how a problem should be represented and which method is best for solving it.

⇒ Production system categories:

- A production system is a method used in Artificial intelligence to solve problems using a set of rules.
- It mainly contains:

1. Rules - Tell what action to perform.
2. working memory - stores the current information.
3. Control Strategy - Decides which rule should be applied.

*Types of production systems.

1. Monotonic production system
2. Non-monotonic production system.
3. Partially commutative production system.

4. Commutative production system.

1. Monotonic production system.

In this system, applying a rule does not prevent other rules from being applied later.

Example:

If we add new information to solve a problem, the old information remains unchanged.

2. Non-monotonic production system.

In this system, applying a rule may change or remove previous information, which can affect other rules.

Example:

A decision made earlier can be changed when new information is received

3. Partially Commutative Production System.

In this system, rules can be applied in different orders, but the final result will still be correct.

Example:

whether Rule A is applied first or Rule B is applied first, we can still reach the solution.

4. Commutative Production System.

In this system, the order of applying rules does not affect the final result.

Example:

Applying Rule A → Rule B gives the same result as Rule B → Rule A.

Type	Main idea
Monotonic	old information is not changed
Non-monotonic	old information can be changed.
Partially commutative	Some rule orders can be changed.
commutative	Rule order does not affect the result.